Chapter 2

Vectors

2.1 What is a Vector?

For most purposes economists think of vectors as an array of real numbers $(x_1, x_2.....x_n)$. The set of all n vectors is written as $\mathcal{R}^n$, pronounced "R n". You have to distinguish between row and column vectors when doing matrix and vector algebra. In this book all vectors are column vectors.

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

The corresponding row vector is written as:

$$\mathbf{x}' = (x_1, x_2.....x_n).$$

The vector $\mathbf{x}'$ is sometimes pronounced as "x prime". Pure mathematicians work with a more abstract definition of a vector and a vector space. Physicists think of vectors as things that have both size and direction, for example velocity and force.

2.2 Vector Addition and Multiplication

2.2.1 Vector Addition

If $\mathbf{x}$ and $\mathbf{y}$ are two vectors in $\mathcal{R}^n$

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}.$$

Figure 2.1 illustrates vector addition with the vectors $\mathbf{x}$ and $\mathbf{y}$ shown nose to tail. The alternative is to show vector addition as a parallelogram as in Figure 2.2.

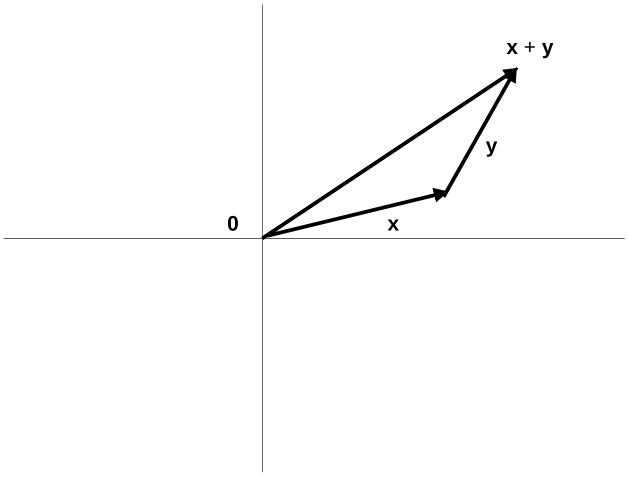

Figure 2.1: Vector Addition

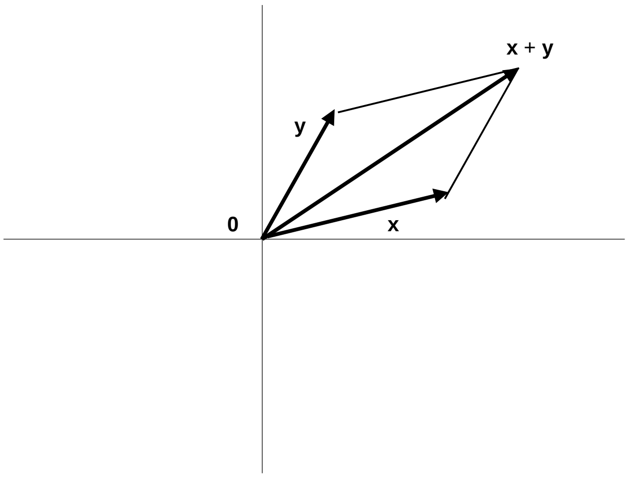

Figure 2.2: Vector Addition Illustrated with a Parallelogram

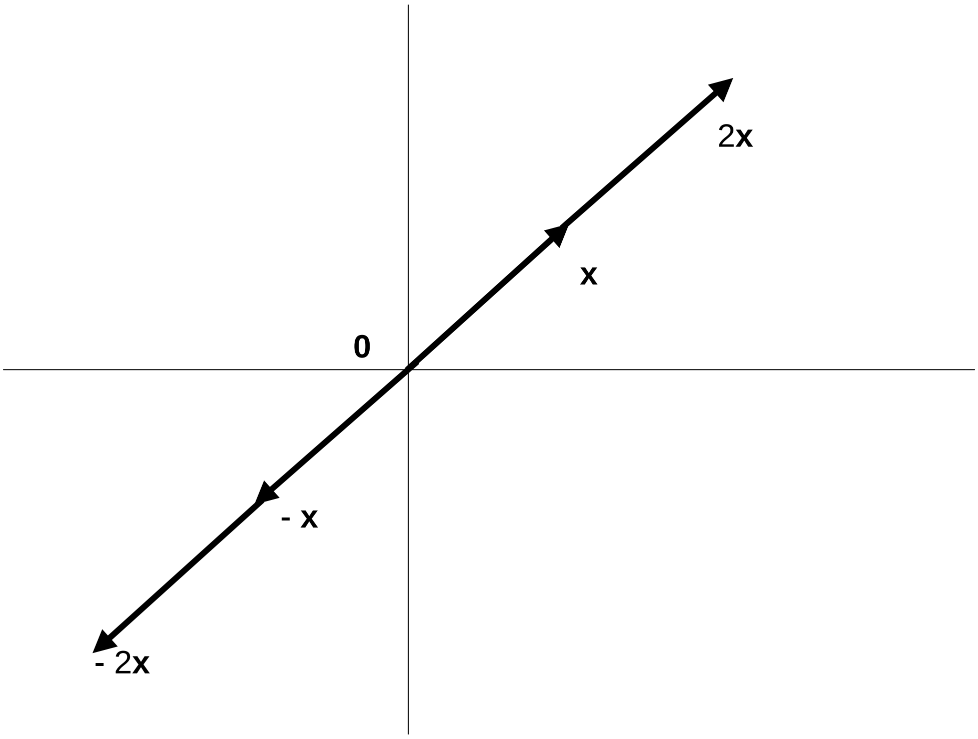

Figure 2.3: Multiplying a Vector by a Scalar

Vector addition is commutative, that is $\mathbf{x}+\mathbf{y}=\mathbf{y}+\mathbf{x}$. This is because

$$\mathbf{x}+\mathbf{y}=\begin{pmatrix} x_1+y_1 \\ x_2+y_2 \\ . \\ . \\ x_n+y_n \end{pmatrix} = \begin{pmatrix} y_1+x_1 \\ y_2+x_2 \\ . \\ . \\ y_n+x_n \end{pmatrix} = \mathbf{y}+\mathbf{x}$$

If $\mathbf{x} \in \mathcal{R}^n$ and $\mathbf{y} \in \mathcal{R}^m$ and n and m are different you cannot add $\mathbf{x}$ and $\mathbf{y}$. In mathematical language the sum of $\mathbf{x}$ and $\mathbf{y}$ is undefined.

2.2.2 Scalar Multiplication

If k is a real number (called a scalar in this context)

$$k\mathbf{x}=k\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} kx_1 \\ kx_2 \\ \vdots \\ kx_n \end{pmatrix} = \mathbf{x}k.$$

Figure 2.3 illustrates a vector $\mathbf{x}$ multiplied by 2, -1 and -2. Multiplying a vector by a positive number keeps it pointing in the same direction but makes it longer or shorter. Multiplying a vector by a negative number makes it point in the opposite direction.

Scalar multiplication is distributive over vector addition, that is

$$k(\mathbf{x}+\mathbf{y}) = k\mathbf{x}+k\mathbf{y} \text{ for all } n \text{ vectors } \mathbf{x} \text{ and } \mathbf{y}.$$

To see why observe that

$$\begin{aligned} k\left(\mathbf{x}+\mathbf{y}\right) &= k\begin{pmatrix} x_1+y_1 \\ x_2+y_2 \\ . \\ . \\ x_n+y_n \end{pmatrix} = \begin{pmatrix} k\left(x_1+y_1\right) \\ k\left(x_2+y_2\right) \\ . \\ . \\ k\left(x_n+y_n\right) \end{pmatrix} \\ &= k\begin{pmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{pmatrix} + k\begin{pmatrix} y_1 \\ y_2 \\ . \\ . \\ y_n \end{pmatrix} = k\mathbf{x}+k\mathbf{y}. \end{aligned}$$

The definition of vector addition also implies that if h and k are scalars and $\mathbf{x}$ a vector

$$\left(h+k\right)\mathbf{x} = h\mathbf{x}+k\mathbf{x}$$

that is scalar multiplication is distributive over scalar addition.

2.2.3 Inner product

The inner product of two vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathcal{R}^n$ is

$$\mathbf{x}'\mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

The inner product is sometime called the "scalar product" or "dot product" and written as $\mathbf{x}\cdot\mathbf{y}$. The properties of the inner product are:

1. The inner product is commutative, that is

   $$\mathbf{x}'\mathbf{y} = \mathbf{y}'\mathbf{x}$$

   for all n vectors $\mathbf{x}$ and $\mathbf{y}$. This is because $\mathbf{x}'\mathbf{y} = \sum_{i=1}^{n} x_i y_i = \sum_{i=1}^{n} y_i x_i = \mathbf{y}'\mathbf{x}$.

2. If $\mathbf{x}$ and $\mathbf{y}$ are n vectors and k is a scalar

   $$k(\mathbf{x}'\mathbf{y}) = (k\mathbf{x})'\mathbf{y} = \mathbf{x}'\left(k\mathbf{y}\right)$$

   because $k(\mathbf{x}'\mathbf{y}) = k\left(\sum_{i=1}^{n} x_i y_i\right) = \sum_{i=1}^{n}(kx_i)y_i = (k\mathbf{x})'\mathbf{y} = \sum_{i=1}^{n} x_i\left(ky_i\right) = \mathbf{x}'\left(k\mathbf{y}\right)$.

3. If $\mathbf{t}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are n vectors

   $$\mathbf{t}'\left(\mathbf{u}+\mathbf{v}\right) = \left(\mathbf{u}+\mathbf{v}\right)'\mathbf{t} = \mathbf{t}'\mathbf{u}+\mathbf{t}'\mathbf{v}$$

   because $\mathbf{t}'\left(\mathbf{u}+\mathbf{v}\right) = \left(\mathbf{u}+\mathbf{v}\right)'\mathbf{t} = \sum_{i=1}^{n} t_i\left(u_i+v_i\right) = \sum_{i=1}^{n} t_i u_i + \sum_{i=1}^{n} t_i v_i = \mathbf{t}'\mathbf{u}+\mathbf{t}'\mathbf{v}$.

4. Also
$$(\mathbf{t}+\mathbf{w})'(\mathbf{u}+\mathbf{v}) = \mathbf{t}'\mathbf{u}+\mathbf{w}'\mathbf{u}+\mathbf{t}'\mathbf{v}+\mathbf{w}'\mathbf{v} = \mathbf{u}'\mathbf{t}+\mathbf{u}'\mathbf{w}+\mathbf{v}'\mathbf{t}+\mathbf{v}'\mathbf{w}$$
because $(\mathbf{t}+\mathbf{w})'(\mathbf{u}+\mathbf{v}) = \sum_{i=1}^{n}(t_i+w_i)(u_i+v_i) = \sum_{i=1}^{n}(t_iu_i+w_iu_i+t_iv_i+w_iv_i)$.

5. Similarly If $\mathbf{t}$, $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are n vectors and a, b, c, d are scalars
$$\begin{aligned}(a\mathbf{t}+b\mathbf{w})'(c\mathbf{u}+d\mathbf{v}) &= (a\mathbf{t})'(c\mathbf{u}+d\mathbf{v})+(b\mathbf{w})'(c\mathbf{u}+d\mathbf{v}) \\ &= (a\mathbf{t})'(c\mathbf{u})+(a\mathbf{t})'(d\mathbf{v})+(b\mathbf{w})'(c\mathbf{u})+(b\mathbf{w})'(d\mathbf{v}) \\ &= ac\mathbf{t}'\mathbf{u}+ad\mathbf{t}'\mathbf{v}+bc\mathbf{w}'\mathbf{u}+bd\mathbf{w}'\mathbf{v}. \qquad (2.1)\end{aligned}$$

2.3 Length and Norm

2.3.1 Definition

For all n vectors $\mathbf{x}$, $\mathbf{x}'\mathbf{x} = \sum_{i=1}^{n} x_i^2$. As $x_i^2 \geq 0$ for all x_i and $x_i^2 = 0$ if and only if $x_i = 0$,
$$\mathbf{x}'\mathbf{x} \geq 0 \text{ for all } \mathbf{x} \text{ and } \mathbf{x}'\mathbf{x} = 0 \text{ if and only if } \mathbf{x} = \mathbf{0}$$
where $\mathbf{0}$ is an n vector with every element 0. Define $\|\mathbf{x}\|$ as
$$\|\mathbf{x}\| = (\mathbf{x}'\mathbf{x})^{\frac{1}{2}}.$$
so $\|\mathbf{x}\| \geq 0$ and $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$. Pythagoras' Theorem implies that in 2 and 3 dimensions $\|\mathbf{x}\|$ is the length of $\mathbf{x}$. Pure mathematicians call $\|\mathbf{x}\|$ the norm of $\mathbf{x}$. It is also called the length of $\mathbf{x}$.

2.3.2 Properties of the Length $\|\mathbf{x}\|$

These are things you need to remember. The first property follows directly from the definition. The others are proved in sections 2.5 and 2.6 of this chapter.

1. If r is a scalar $\|r\mathbf{x}\| = |r|\,\|\mathbf{x}\|$ where $|r|$ is the absolute value of r, so $|r| = r$ if $r \geq 0$ and $|r| = -r$ if $r < 0$.

2. $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\|\,\|\mathbf{y}\|\cos\theta$ where θ is the angle between $\mathbf{x}$ and $\mathbf{y}$.

3. Cauchy-Schwarz inequality:
$$|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\|\,\|\mathbf{y}\|.$$

4. Triangle inequality
$$\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\|+\|\mathbf{y}\|$$
$$\|\mathbf{x}-\mathbf{y}\| \leq \|\mathbf{x}\|+\|\mathbf{y}\|.$$

5. If $\mathbf{x}'\mathbf{y} = 0$, $\mathbf{x} \neq 0$ and $\mathbf{y} \neq 0$ the angle between $\mathbf{x}$ and $\mathbf{y}$ is $90°$ because $\cos 90° = 0$. The vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be orthogonal.

6. If $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\|\,\|\mathbf{y}\|$, $\mathbf{x}$ and $\mathbf{y}$ are parallel and point in the same direction because $\cos 0 = 1$.

7. If $\mathbf{x}'\mathbf{y} = -\|\mathbf{x}\|\,\|\mathbf{y}\|$, $\mathbf{x}$ and $\mathbf{y}$ are parallel and point in opposite directions because $\cos 180° = -1$.

2.4 The Least Squares Problem

2.4.1 The Least Squares Problem With Vector Notation

You will become very familiar with least squares problems in econometrics. This section of notes covers the simplest case. It also generates inequality 2.3 which is an essential step on the way to the Cauchy-Schwarz inequality. The problem here is finding the value of b that minimizes

$$L(b, \mathbf{x}, \mathbf{y}) = \sum_{t=1}^{T} (y_t - bx_t)^2.$$

You will come to think of this as the regression of y onto a single variable x without an intercept. Suppose $\mathbf{x}$ and $\mathbf{y}$ are T vectors. Given the definition of the norm $\|\mathbf{y} - b\mathbf{x}\|$

$$L(b, \mathbf{x}, \mathbf{y}) = \sum_{t=1}^{T} (y_t - bx_t)^2 = \|\mathbf{y} - b\mathbf{x}\|^2 = (\mathbf{y} - b\mathbf{x})'(\mathbf{y} - b\mathbf{x})$$

Expanding the brackets using the result in equation 2.1 on inner products gives

$$\begin{aligned} L(b, \mathbf{x}, \mathbf{y}) &= (\mathbf{y} - b\mathbf{x})'(\mathbf{y} - b\mathbf{x}) \\ &= \mathbf{y}'\mathbf{y} - b\mathbf{x}'\mathbf{y} - b\mathbf{y}'\mathbf{x} + b^2\mathbf{x}'\mathbf{x} \\ &= \mathbf{y}'\mathbf{y} - 2b\mathbf{x}'\mathbf{y} + b^2\mathbf{x}'\mathbf{x} \end{aligned}$$

because $\mathbf{x}'\mathbf{y} = \mathbf{x}'\mathbf{y}$.

Assume that $x_t \neq 0$ for some t so $\mathbf{x} \neq 0$. This implies that $\mathbf{x}'\mathbf{x} > 0$. Then completing the square

$$\begin{aligned} L(b, \mathbf{x}, \mathbf{y}) &= \mathbf{y}'\mathbf{y} - 2b\mathbf{x}'\mathbf{y} + b^2\mathbf{x}'\mathbf{x} \qquad (2.2) \\ &= \mathbf{x}'\mathbf{x}\left(b - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})\right)^2 + \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2 \\ &\geq \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2 \end{aligned}$$

for all b and

$$L(b, \mathbf{x}, \mathbf{y}) = \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2$$

if and only if $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$. Thus $L(b, \mathbf{x}, \mathbf{y})$ is minimized by setting

$$b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$$

and has a minimum value of

$$\mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2.$$

Recall that $L(b, \mathbf{x}, \mathbf{y})$ is a sum of squares, so $L(b, \mathbf{x}, \mathbf{y}) \geq 0$ for all b including $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$ at which point $L(b, \mathbf{x}, \mathbf{y}) = \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2$. Thus $\mathbf{y}'\mathbf{y} \geq (\mathbf{x}'\mathbf{x})^{-\mathbf{1}}(\mathbf{x}'\mathbf{y})^{\mathbf{2}}$. Recall that by assumption $\mathbf{x} \neq 0$ so $\mathbf{x}'\mathbf{x} > 0$. Multiplying by $\mathbf{x}'\mathbf{x}$ and rearranging gives

$$(\mathbf{x}'\mathbf{y})^2 \leq (\mathbf{x}'\mathbf{x})(\mathbf{y}'\mathbf{y}). \qquad (2.3)$$

2.4.2 The Least Squares Problem Without Vector Notation

You can also solve this simple least squares problem without using vectors. The problem is to find b that minimizes the sum of squares.

$$\sum_{t=1}^{n} (y_t - bx_t)^2 .$$

Expanding the brackets

$$\begin{aligned}\sum_{t=1}^{n} (y_t - bx_t)^2 &= \sum_{t=1}^{n} \left(y_t^2 - 2by_tx_t + b^2x_t^2\right) \\ &= \sum_{t=1}^{n} y_t^2 - 2b\left(\sum_{t=1}^{n} x_ty_t\right) + b^2\left(\sum_{t=1}^{n} x_t^2\right).\end{aligned}$$

Assume that at least one of the $x_t's$ is not zero, so $\sum_{t=1}^{n} x_t^2 > 0$. Completing the square

$$\begin{aligned}\sum_{t=1}^{n} (y_t - bx_t)^2 &= \left(\sum_{t=1}^{n} x_t^2\right)\left(b - \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right)\right)^2 \\ &\quad + \sum_{t=1}^{n} y_t^2 - \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right)^2\end{aligned}$$

which is minimized by setting

$$b = \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right) = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y}).$$

and has a minimum value of

$$\sum_{t=1}^{n} y_t^2 - \left(\sum_{t=1}^{n} x_t^2\right)^{-1}\left(\sum_{t=1}^{n} x_ty_t\right)^2 = \mathbf{y}'\mathbf{y} - (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})^2 .$$

Notice that expressions like $\sum_{t=1}^{n} x_ty_t$ take longer to write down and are harder to work with than the corresponding vector expression $\mathbf{x}'\mathbf{y}$. The algebra becomes even worse if you are working with the general least squares problem of minimizing

$$\sum_{t=1}^{n}\left(y_t - \sum_{k=1}^{k} x_{tk}b_k\right)^2 .$$

If you are to survive your econometrics course you must make the switch from the notation $\sum_{t=1}^{n} x_ty_t$ to vector notation $\mathbf{x}'\mathbf{y}$ as rapidly as possible.

2.5 Inequalities for Vectors

2.5.1 The Cauchy-Schwarz Inequality

The Cauchy-Schwarz inequality says that for any vectors $\mathbf{x}$ and $\mathbf{y}$ with the same number of elements

$$|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\| \,. \tag{2.4}$$

Here $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$ are the norm or length of $\mathbf{x}$ and $\mathbf{y}$, so $\|\mathbf{x}\|^2 = \mathbf{x}'\mathbf{x}$ and $\|\mathbf{y}\|^2 = \mathbf{y}'\mathbf{y}$. The expression $|\mathbf{x}'\mathbf{y}|$ is the absolute value of $\mathbf{x}'\mathbf{y}$. (The absolute value of a number p is p if $p \geq 0$ and $-p$ if $p < 0$. Note that this implies that $p^2 = |p|^2$)

The Cauchy-Schwarz inequality is derived from inequality 2.3 which states that $(\mathbf{x}'\mathbf{y})^2 \leq (\mathbf{x}'\mathbf{x})(\mathbf{y}'\mathbf{y})$. From the definitions of absolute value and norm $|\mathbf{x}'\mathbf{y}|^2 = (\mathbf{x}'\mathbf{y})^2$, $(\mathbf{x}'\mathbf{x}) = \|\mathbf{x}\|^2$ and $\mathbf{y}'\mathbf{y} = \|\mathbf{y}\|^2$ so the inequality $(\mathbf{x}'\mathbf{y})^2 \leq (\mathbf{x}'\mathbf{x})(\mathbf{y}'\mathbf{y})$ can be written as

$$|\mathbf{x}'\mathbf{y}|^2 \leq \|\mathbf{x}\|^2 \, \|\mathbf{y}\|^2 \tag{2.5}$$

which implies the Cauchy-Schwarz inequality $|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|$. Inequality 2.3 was proved under the assumption that $\mathbf{x} \neq 0$, but as both sides of the Cauchy-Schwarz inequality are zero when $\mathbf{x} = 0$ it also holds when $\mathbf{x} = 0$.

2.5.2 The Triangle Inequality

The triangle inequality states that

$$\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| \,.$$

Geometrically you can think of $\mathbf{x}+\mathbf{y}$, $\mathbf{x}$ and $\mathbf{y}$ as three sides of a triangle. The inequality simply states that one side of a triangle cannot be longer than the sum of the other two sides, as illustrated in Figure 2.4.

The triangle inequality is a direct consequence of the Cauchy-Schwarz inequality. From the definition of $\|\mathbf{x}+\mathbf{y}\|$

$$\|\mathbf{x}+\mathbf{y}\|^2 = (\mathbf{x}+\mathbf{y})'(\mathbf{x}+\mathbf{y}) = \mathbf{x}'\mathbf{x} + \mathbf{2}\mathbf{x}'\mathbf{y} + \mathbf{y}'\mathbf{y}$$

From the Cauchy-Schwarz inequality $|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|$. As $z \leq |z|$ for any number z this implies that $\mathbf{x}'\mathbf{y} \leq \|\mathbf{x}\| \, \|\mathbf{y}\|$ so

$$\begin{aligned} \|\mathbf{x}+\mathbf{y}\|^2 &= \mathbf{x}'\mathbf{x} + \mathbf{2}\mathbf{x}'\mathbf{y} + \mathbf{y}'\mathbf{y} \\ &\leq \|\mathbf{x}\|^2 + \mathbf{2}\,\|\mathbf{x}\| \, \|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 \,. \end{aligned}$$

As $\|\mathbf{x}+\mathbf{y}\|$, $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$ are all non negative this implies the triangle inequality $\|\mathbf{x}+\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$. Replacing $\mathbf{y}$ by $-\mathbf{y}$ and noting that $\|\mathbf{y}\| = \|-\mathbf{y}\|$ the triangle inequality implies that

$$\|\mathbf{x}-\mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\| \,.$$

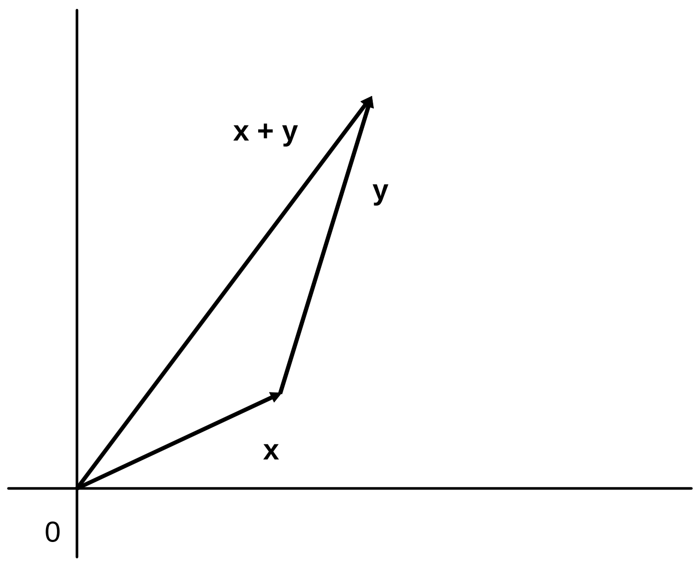

Figure 2.4: triangle inequality

2.6 The Angle Between Two Vectors

2.6.1 Showing That $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos\theta$

Think of $\mathbf{x}$ and $\mathbf{y}$ as non zero vectors and b as a scalar. In Figure 2.5 $b\mathbf{x}$ is a scalar multiple of the vector $\mathbf{x}$. The least squares problem is finding b to minimize $\|\mathbf{y}-b\mathbf{x}\|$ which is the distance between the point C at the end of the vector $\mathbf{y}$ and the horizontal straight line from the origin 0 that goes through the end of the vector $\mathbf{x}$. Geometrically you find the point $b\mathbf{x}$ on the line $0\mathbf{x}$ that minimizes the length of $\mathbf{y}-b\mathbf{x}$ by drawing the line from C to B that is at 90° to the vector $\mathbf{x}$..

As $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$ solves the least squares minimization problem the length of the line OB is the length of the vector $b\mathbf{x}$ where $b = (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y})$. Think about the case where b is positive as shown in Figure 2.5. As $\|\mathbf{x}\| = (\mathbf{x}'\mathbf{x})^{\frac{1}{2}}$ the length of OB is $\|\mathbf{x}\| \, b = \|\mathbf{x}\| (\mathbf{x}'\mathbf{x})^{-1}(\mathbf{x}'\mathbf{y}) = (\mathbf{x}'\mathbf{x})^{-\frac{1}{2}}(\mathbf{x}'\mathbf{y})$. Elementary trigonometry and Figure 2.5 imply that if θ is the angle between the vectors $\mathbf{x}$ and $\mathbf{y}$

$$\cos\theta = \frac{0B}{0C} = \frac{(\mathbf{x}'\mathbf{x})^{-\frac{1}{2}}(\mathbf{x}'\mathbf{y})}{\|\mathbf{y}\|} = \frac{\mathbf{x}'\mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|} \tag{2.6}$$

where $\cos\theta$ is the cosine of the angle θ. Equation 2.6 implies that

$$\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos\theta.$$

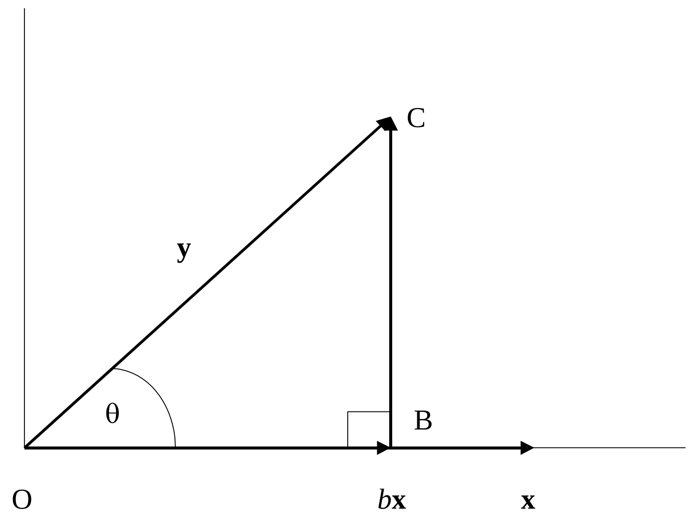

Figure 2.5:

In words this says that the inner product of the vectors $\mathbf{x}$ and $\mathbf{y}$ is the product of their lengths and the cosine of the angle between the two vectors. The relationship also holds when b is negative, so the angle between the two vectors is more than 90° and, as the graph of $\cos\theta$ in Figure 2.6 shows, $\cos\theta < 0$. One of the properties of the cosine is that for all θ with $0^\circ \leq \theta \leq 360^\circ$

$$\cos\theta = \cos(360 - \theta).$$

This has the useful implication that it does not matter which way round you think of the angle between the two vectors, as shown in Figure 2.7 . If you think of the angle between the two vectors as the smaller of the two angles between the two vectors, so it lies between 0° and 180°. The graph of the function $\cos\theta$ in Figure 2.6 shows that

$$0 < \cos\theta < 1 \text{ if } 0 < \theta < 90^\circ$$

and

$$-1 < \cos\theta < 0 \text{ if } 90^\circ < \theta < 180^\circ$$

which gives a result sufficiently useful to be stated formally.

Proposition 1

- $\mathbf{x}'\mathbf{y} > 0$ *if the angle between* $\mathbf{x}$ *and* $\mathbf{y}$ *is between* 0 *and* 90°.
- $\mathbf{x}'\mathbf{y} < 0$ *if the angle between* $\mathbf{x}$ *and* $\mathbf{y}$ *is between* 90° *and* 180°.

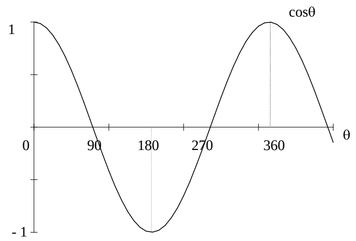

Figure 2.6: The Graph of $\cos\theta$

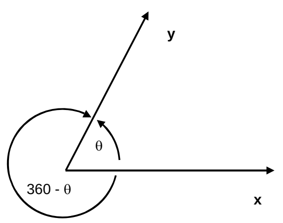

Figure 2.7:

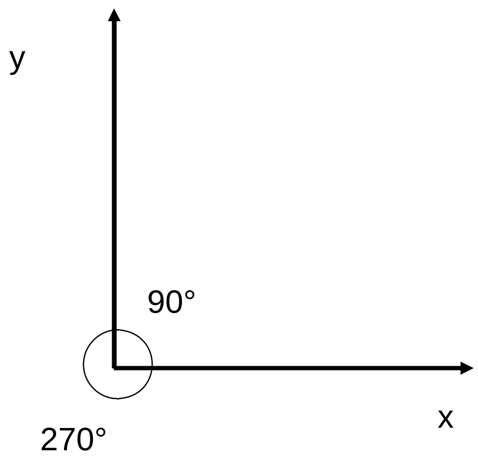

Figure 2.8:

2.6.2 Orthogonal Vectors

Suppose $\mathbf{x}$ and $\mathbf{y}$ are non zero vectors, so $\|\mathbf{x}\| > 0$ and $\|\mathbf{y}\| > 0$. Suppose also that $\mathbf{x}'\mathbf{y} = 0$. This implies that $\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta = 0$ where θ is the angle between the vectors $\mathbf{x}$ and $\mathbf{y}$. As $\|\mathbf{x}\| > 0$ and $\|\mathbf{y}\| > 0$ this implies that $\cos\theta = 0$. Given the graph of $\cos\theta$ this implies that $\theta = 90°$ or $\theta = 270°$.

As Figure 2.8 shows you can measure the angle between $\mathbf{x}$ and $\mathbf{y}$ as 90° or 270° . It is convenient to choose the smaller angle 90°. In mathematical language the vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be orthogonal if $\mathbf{x}'\mathbf{y} = 0$ so the vectors are at an angle of 90° to each other. In everyday English an angle of 90° is called a "right angle".

2.6.3 Parallel vectors

If

$$\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta = \|\mathbf{x}\| \|\mathbf{y}\|$$

then $\cos\theta = 1$ so $\theta = 0$. The angle between $\mathbf{x}$ and $\mathbf{y}$ is 0°. The two vectors are parallel and point in the same direction.

If

$$\mathbf{x}'\mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos\theta = - \|\mathbf{x}\| \|\mathbf{y}\|$$

then $\cos\theta = -1$ so $\theta = 180°$. The two vectors are parallel and point in opposite directions.

2.7 Vectors, Lines, Planes and Hyperplanes

2.7.1 Vectors and Lines in $\mathcal{R}^2$

The objective of this section is to introduce the idea of a hyperplane, which is a generalization of a line in two dimensional space, and a plane in three dimensional space. Hyperplanes are important for an intuitive understanding of partial derivatives, and also for the theory of concavity and convexity that underlies optimization theory.

One of microeconomics' favorite equations is the budget line; with two goods x_1 and x_2, with prices p_1 and p_2 this is the set of points satisfying

$$p_1x_1 + p_2x_2 = m \tag{2.7}$$

where m is the amount the consumer has to spend. This is straight line; if $p_2 \neq 0$ equation 2.7 can be written as

$$x_2 = \frac{m}{p_2} - \frac{p_1}{p_2}x_1$$

so the slope of the budget line is $-\frac{p_1}{p_2}$. If $p_2 = 0$ the line is vertical. In vector notation equation 2.7 can be written as

$$\mathbf{p}'\mathbf{x} = m.$$

Another way at looking at this equation is to choose any point $\mathbf{x}_0$ on the line so $\mathbf{p}'\mathbf{x}_0{=}m$ and write the equation as $\mathbf{p}'\mathbf{x} = \mathbf{p}'\mathbf{x}_0$, or rearranging

$$\mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = 0.$$

Recall that the inner product $\mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = \|\mathbf{p}\| \|\mathbf{x} - \mathbf{x}_0\| \cos\theta$ where θ is the angle between the two vectors $\mathbf{p}$ and $\mathbf{x} - \mathbf{x}_0$. Thus as $\cos 90° = 0$, if the inner product is 0, the angle is 90° and the two vectors are said to be "orthogonal". The line, that is the set of points satisfying $\mathbf{p}'(\mathbf{x} - \mathbf{x}_0)$, is precisely the set of points for which $\mathbf{x} - \mathbf{x}_0$ is orthogonal to $\mathbf{p}$. This is illustrated in Figure 2.9. The vector $\mathbf{p}$ orthogonal to the line is called the **normal vector**.

2.7.2 Vectors and Planes in $\mathcal{R}^3$

With three goods the budget equation becomes

$$p_1x_1 + p_2x_2 + p_3x_3 = m.$$

The vector notation appears unchanged as $\mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = 0$, but now you have to think of the vectors as living in three dimensional space $\mathcal{R}^3$ rather than two dimensional space $\mathcal{R}^2$. Geometrically, in three dimensional space, given a fixed vector $\mathbf{p}$, the set of vectors orthogonal to $\mathbf{p}$ is a plane. The vector $\mathbf{p}$ is again called the **normal vector** to the plane.

You can perhaps imagine the plane and its normal vector $\mathbf{p}$ by thinking of your pen as $\mathbf{p}$. If you put one end of the pen on the table top, and point the pen

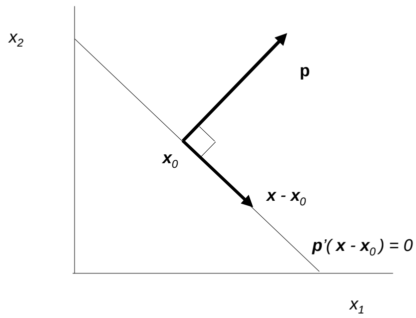

Figure 2.9: The vector $\mathbf{p}$ is orthogonal to the line $\mathbf{p}'(\mathbf{x} - \mathbf{x_0}) = 0$.

vertically upwards, all the vectors in the plane of the table top are horizontal, so are orthogonal to the pen. It is possible to lift up a corner of the table so the table top is no longer horizontal, whilst keeping the angle between pen and table top fixed at 90°, so the vector is still orthogonal to the plane. (Take your coffee mug off the table before trying this.)

2.7.3 Vectors and Hyperplanes in $\mathcal{R}^n$

With n goods the budget equation becomes

$$p_1x_1 + p_2x_2 + ... + p_nx_n = m.$$

Again the budget equation can be written as $\mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = 0$, but $\mathbf{p}$ and $\mathbf{x} - \mathbf{x}_0$ are now n vectors. A hyperplane is defined as follows:

Definition 2 *A hyperplane in $\mathcal{R}^n$ is a set of the form*

$$\{\mathbf{x} : \mathbf{x} \in \mathcal{R}^n, \mathbf{p}'(\mathbf{x} - \mathbf{x}_0) = 0\}$$

where $\mathbf{p}$ is a non-zero vector in $\mathcal{R}^n$, and $\mathbf{x}_0$ is a vector in $\mathcal{R}^n$.

In two dimensional space a hyperplane is a straight line, and in three dimensional space a hyperplane is a plane. For any n all the vectors $\mathbf{x} - \mathbf{x}_0$ are orthogonal to $\mathbf{p}$, and $\mathbf{p}$ is still called the normal vector.